

Aryan Maheshwari

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EDUCATION

University of Wisconsin-Madison

Madison, WI

Bachelor of Science in Computer Science & Data Science — GPA: 3.7/4

Sept 2023 – May 2027

- **Coursework:** Data Structures & Algorithms, OS, Big Data Systems, Database Management & Systems

EXPERIENCE

Software Engineering Intern

May 2025 – Dec 2025, May 2026 – Current

HG Insights

Santa Barbara, CA

- Designed and built the backend pipeline in **Python** for a Revenue Growth & Market Analysis model, architecting a **LangGraph** DAG that orchestrates SEC filing ingestion and a trained **SLM** - deployed on **AWS Agentcore, Lambda, ECR (Docker)**.
- Drove data curation and normalization that boosted data completeness for **15M+** unique rows from **20% to 100%**, then categorized them into a 180-class taxonomy to make the training set consistent.
- Shipped the model into production, powering market analysis for **75% of the tech companies in the Fortune 100**.

Undergraduate Research Assistant

Sep. 2025 – May 2026

University of Wisconsin-Madison

Madison, WI

- Built a **ML pipeline** to classify natural speech from raw EEG signals, implementing the methodology from Berg et al. (2021) IEEE Paper in Python end-to-end - signal preprocessing and classification.
- Trained and evaluated multiple **neural network** architectures to improve classification, reaching **81% accuracy** on recognizing critical Mandarin phonetic contrasts.
- Productionized an **MCP server** exposing the model via the local network, giving lab students a clean interface to run the neural network and classify Mandarin phonetics without any ML setup through Claude Code.

Product Intern

May 2024 - August 2024

Aeries Technology

Mumbai, India

- Built a full-stack candidate-ranking tool with a **React** frontend and an agentic Python backend, cutting manual screening of hundreds of applicants from weeks to minutes.
- Developed **REST APIs** connecting the front end to the ranking service, with prompt-injection checks and output verification to keep rankings reliable.
- Containerized with **Docker** and deployed on **GCP**, running end-to-end from resume input to a ranked shortlist surfaced to HR

PROJECTS

Wisconsin Racing

Sept 2025 - Current

- **Led** the design for the steering-wheel firmware and low-level control software in **C** on an ESP32, managing real-time driver inputs over CAN bus and analog potentiometers
- Built and programmed aerodynamic data-acquisition tools to record on-track performance, validated on the car at Ford's Wind Tunnel in Detroit, MI
- Placed **1st** in Design, **2nd** in Autocross, and **7th Overall** at Formula FSAE Michigan 2025

AI-Powered Mock Interview System

January 2025 – Present

- Architected a multi-agent interview platform, coordinating specialized agents for resume analysis, question generation, and response evaluation, served through a **FastAPI** backend and **React** frontend
- Reached **150+ users** in the first month during recruiting season, running voice-based mock interviews end-to-end from resume upload to scored feedback
- Engineered multi-model routing across hosted (OpenAI) and self-hosted (Ollama) LLMs, decoupling the system from any single provider for cost and availability control

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, SQL (Postgres), HTML/CSS, R

Frameworks & Libraries: FastAPI, React, Node.js, Flask, LangGraph, PyTorch, pandas, NumPy

Cloud & Infrastructure: AWS (Lambda, ECR, Bedrock, AgentCore), GCP, Docker, Databricks, CI/CD

Contributions & Achievements: Patent Filed for Novel Data Structure, Open Source Contribution to VSCode